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Tab arrangement for retaining support elements of substrate support

Abstract

Embodiments of substrate supports are provided herein. In some embodiments, a substrate support includes: a first plate having a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings; a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate; and a plurality of support elements disposed in each of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings.

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Background/Summary

FIELD

(1) Embodiments of the present disclosure generally relate to a substrate processing equipment.

BACKGROUND

(2) Substrate processing systems typically include process chambers for performing a desired process, such as a degas process, on one or more substrates disposed therein. Typically, semiconductor substrates are degassed in a dedicated degas process module prior to a subsequent process step such as deposition as part of a standard process flow to outgas unwanted materials and contaminant prior to further processing. A degas chamber may include a heated substrate support. The heated substrate support may provide one or more of a controlled sealing gap, reduced metal contamination, and reduced thermal conduction to the substrate. To achieve such features, the heated substrate support may include a heater plate having support elements disposed in the heater plate for elevating the substrate and supporting the substrate with minimal contact with a remainder of the heated substrate support.

(3) The support elements may be disposed in a pocket or opening in the heated substrate support. However, conventional means for retaining the support elements in the pocket or opening may put too much pressure or force on the support elements when swaged, leading to cracking or breaking of the support elements.

(4) Accordingly, the inventors have provided embodiments of improved substrate supports.

SUMMARY

(5) Embodiments of substrate supports are provided herein. In some embodiments, a substrate support includes: a first plate having a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings; a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate; and a plurality of support elements disposed in each of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings.

(6) In some embodiments, a substrate support includes: a first plate comprising: a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings; a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate; and a plurality of support elements disposed in each of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings; a second plate coupled to a lower surface of the first plate, the second plate having one or more heater elements; and a support shaft coupled to the first plate and the second plate.

(7) In some embodiments, a degas chamber includes: a chamber body defining an interior volume therein; a substrate support for supporting a substrate disposed in the interior volume, the substrate support comprising: a pedestal having a first plate having a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings, a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate, and a plurality of support elements disposed in each of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings; and a support shaft coupled to the pedestal; and a heating lamp disposed in the interior volume opposite the substrate support.

(8) Other and further embodiments of the present disclosure are described below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Embodiments of the present disclosure, briefly summarized above and discussed in greater detail below, can be understood by reference to the illustrative embodiments of the disclosure depicted in the appended drawings. However, the appended drawings illustrate only typical embodiments of the disclosure and are therefore not to be considered limiting of scope, for the disclosure may admit to other equally effective embodiments.

(2) FIG. **1** depicts a schematic side view of a degas chamber in accordance with at least some embodiments of the present disclosure.

(3) FIG. **2** depicts a schematic cross-sectional view of a substrate support in accordance with at least some embodiments of the present disclosure.

(4) FIG. **3** depicts an isometric view of a plurality of tabs in accordance with at least some embodiments of the present disclosure.

(5) FIG. **4** depicts an isometric view of a plurality of tabs in accordance with at least some embodiments of the present disclosure.

(6) FIG. **5**A depicts a cross-sectional side view of a tab in accordance with at least some embodiments of the present disclosure.

(7) FIG. **5**B depicts a cross-sectional side view of a tab in accordance with at least some embodiments of the present disclosure.
(8) FIG. **5**C depicts a cross-sectional side view of a tab in accordance with at least some embodiments of the present disclosure.
(9) FIG. **6** depicts a schematic cross-sectional view of a portion of a first plate in accordance with at least some embodiments of the present disclosure.
(10) FIG. **7** depicts a schematic cross-sectional view of a portion of a first plate with a plurality of tabs in a bent position in accordance with at least some embodiments of the present disclosure.
(11) To facilitate understanding, identical reference numerals have been used, where possible, to designate identical elements that are common to the figures. The figures are not drawn to scale and may be simplified for clarity. Elements and features of one embodiment may be beneficially incorporated in other embodiments without further recitation.

DETAILED DESCRIPTION

(12) Embodiments of substrate supports for use in semiconductor process chambers, for example, degas chambers, are provided herein. The substrate supports generally include a plurality of support elements for raising a substrate above an upper surface of the substrate support. The inventors have observed that the plurality of support elements may be prone to cracking or breaking due to forces exerted thereon when fixing or retaining the plurality of support elements to an upper plate of the substrate support. The upper plate, or first plate, provided herein include a plurality of tabs disposed about the plurality of support element such that when swaged to retain the plurality of support elements, the plurality of tabs advantageously exert reduced or no pressure on the plurality of support elements. In some embodiments, the plurality of tabs may comprise more than two tabs so that less force is needed to swage the plurality tabs, thereby exerting less pressure on the plurality of support elements. The bending plane or geometry of the plurality of tabs may also be suitably selected to advantageously exert reduced or no pressure on the plurality of support elements.
(13) FIG. **1** depicts a schematic side view of a degas chamber, or chamber **100**, in accordance with at least some embodiments of the present disclosure. The chamber **100** includes a chamber body **102** having sidewalls **132** and a lid **134** that enclose an interior volume **104** of the chamber **100**. The chamber body **102** may be made of a suitable material, such as aluminum or the like. A substrate support **120** is disposed in the interior volume **104** and configured to support a substrate **112** thereon. The substrate support **120** may include a pedestal **106** coupled to a support shaft **108**. The support shaft **108** may be hollow to accommodate conduits therethrough (e.g., fluid lines, power lines, or the like) that extend to the pedestal **106**. The pedestal **106** can include a heater plate (discussed in more detail in FIG. **2** below).
(14) A heating lamp **110** may be disposed in the interior volume **104** opposite the substrate support **120** and configured to heat the interior volume **104** to perform a degas procedure. The heating lamp **110** may comprise any suitable heating components, for example, one or more heater lamps, heater coils, resistive heater elements, or the like.
(15) In some embodiments, a gas supply **118** is coupled to the chamber body **102** and configured to supply one or more gases to the interior volume **104**. For example, the gas supply **118** may supply an inert gas such as argon gas. The gas supply **118** may also supply one or more suitable process gases. In some embodiments, a pump **122** is coupled to the chamber body **102** and configured to exhaust the interior volume **104**, including outgas materials and process gases. The pump **122** or a second pump (not shown) may regulate a pressure of the interior volume **104**. A slit valve **130** may be coupled to the chamber body **102** and configured for transfer of the substrate **112** into and out of the interior volume **104**.
(16) FIG. **2** depicts a schematic cross-sectional view of a substrate support **120** in accordance with at least some embodiments of the present disclosure. The substrate support **120** generally comprises a first plate **204** having a plurality of openings **226**, or pockets, extending into the first plate **204**

from an upper surface **208** of the first plate **204**. A plurality of support elements **218** are disposed on a bottom surface **260** of the plurality of openings **226**. The plurality of support elements **218** in each of the plurality of openings **226** are configured to raise the substrate **112** above the upper surface **208** and provide a low contact area between the substrate **112** and the pedestal **106**. In some embodiments, the first plate **204** is formed from a process compatible metallic material, for example stainless steel. The first plate **204** may be at least as large as the substrate **112** to be supported thereon. Substrate diameters may be, for example 200, 300, or 450 mm (7.87, 11.81, or 17.72 inches), although larger and smaller diameter substrates may also benefit from the features of the present disclosure.

(17) In some embodiments, a second plate **206** is coupled to a lower surface **264** of the first plate **204**. The second plate **206** may generally include one or more heater elements **228**. In some embodiments, the one or more heater elements **228** comprise one or more inner heater elements **234** disposed in an inner region **232** of the second plate **206** and one or more outer heater elements **244** disposed in an outer region **242** of the second plate **206**. The one or more heater elements **228** may be any suitable heating elements such as, for example, resistive heating elements, lamps, or the like. The one or more heater elements **228** may be coupled to a heater power source **236** for providing power to the one or more heater elements **228**. In some embodiments, the heater power source **236** may provide power to the one or more inner heater elements **234** and the one or more outer heater elements **244**. In some embodiments, the heater power source **236** comprises a first power source for the one or more inner heater elements **234** and a second power source for the one or more outer heater elements **244**.

(18) In some embodiments, the second plate includes a thermal break **224** disposed between the one or more inner heater elements **234** and the one or more outer heater elements **244**. In some embodiments, the thermal break **224** is a gap configured to reduce thermal crosstalk between the inner region **232** and the outer region **242**. In some embodiments, the support shaft **108** is coupled to the first plate **204** and the second plate **206**.

(19) In some embodiments, the plurality of openings **226** are disposed at regular intervals along circles that are concentric with a center **268** of the first plate **204**. In some embodiments, the plurality of support elements **218** may be disposed on the upper surface **208** along a random pattern. In some embodiments, the plurality of support elements **218** are ball-shaped or have other suitable shapes. In some embodiments, the plurality of support elements **218** are made of non-metal or sapphire balls. The plurality of support elements **218** may be complete spheres, have a spherical shape with a flattened bottom, or the like.

(20) The low contact area facilitated by the plurality of support elements **218** may also help reduce or prevent contamination and allows for elevating the substrate **112** to degas a backside of the substrate **112** without having to flip the substrate **112** over. The low contact area between the upper surface **208** and the substrate **112** also reduces or prevents substrate scratching or sticking. The plurality of support elements **218** may stick out very slightly from the upper surface **208** so that there is adequate thermal coupling between the substrate **112** and the first plate **204**. For example, the plurality of support elements **218** may be raised about 0.005 inches to about 0.02 inches from the upper surface **208** of the first plate **204**.

(21) In some embodiments, the first plate **204** may include gas grooves **248** on the upper surface **208** for dispersing a backside gas. The gas grooves **248** may be arranged in any suitable shape. The gas grooves **248** may be coupled to a backside gas source **246**. In some embodiments, the backside gas source **246** may deliver backside gas at a first location proximate the center **268** of the first plate **204**, and the gas grooves **248** may extend radially outward from the first location. The backside gas source **246** may extend to the pedestal **106** through the support shaft **108**.

(22) In some embodiments, the first plate **204** includes an outer peripheral notch **252**. The outer peripheral notch may be configured to support a chamber component, for example, a process kit component such as an edge or focus ring. In some embodiments, an outer lip **262** extends

downward from an outer edge of the first plate **204**. In some embodiments, the outer lip **262** surrounds the second plate **206**. In some embodiments, the pedestal **106** includes a base plate **212** coupled to the first plate **204**, for example, to the outer lip **262** of the first plate **204**. The base plate **212** may be directly coupled to the support shaft **108**. In some embodiments, the first plate **204** includes one or more slots **254** in a peripheral region of the first plate **204**. The one or more slots **254** may comprise alignment features or openings for lift pin assemblies.

(23) FIG. **3** depicts an isometric view of a plurality of tabs **310** in accordance with at least some embodiments of the present disclosure. FIG. **4** depicts an isometric view of a plurality of tabs in accordance with at least some embodiments of the present disclosure. The plurality of tabs **310** comprise three or more tabs disposed in each respective opening of the plurality of openings **226** and extending from a first floor **312** of the plurality of openings **226** towards the upper surface **208** of the first plate **204**. The three or more tabs advantageously require less force to swage the plurality of tabs as compared to less than three tabs for retaining the plurality of support elements **218**. The plurality of tabs **310** are disposed around each of the plurality of support elements **218** and configured to retain the plurality of support elements **218** in the plurality of openings **226**. In some embodiments, the plurality of tabs **310** include 4 or more tabs. In some embodiments, the plurality of tabs **310** consist of 4 tabs as depicted in FIG. **3** or 6 tabs as depicted in FIG. **4**. The plurality of tabs **310** may be arranged at regular intervals about each of the plurality of support elements **218**. In use, the plurality of tabs **310** are swaged to plastically deform inward toward each other to retain the support element disposed therein via any suitable swage tool. In some embodiments, the plurality of tabs **310** are made of a metal. In some embodiments, the plurality of tabs **310** are made of the same material as the first plate **204**.

(24) Each of the plurality of tabs **310** include an inner surface **322** facing one of the plurality of support elements **218** and an outer surface **324** opposite the inner surface **322**. In some embodiments, a surface area of the inner surface **322** is less than a surface area of the outer surface **324**. In some embodiments, the inner surface **322** is curved to conform with the geometry of the one or more support elements **218**.

(25) FIG. **5**A depicts a cross-sectional side view of a tab of the plurality of tabs **310** in accordance with at least some embodiments of the present disclosure. In some embodiments, the bottom surface **260** of each respective opening includes the first floor **312** and a second floor **510** extending from the first floor **312** towards the lower surface **264** of the first plate **204**. The second floor **510** is configured to accommodate the plurality of support elements **218** such that the plurality of support elements **218** may rest on the second floor **510**. In some embodiments, the second floor **510** is rounded corresponding with a shape of the plurality of support elements **218**.

(26) The plurality of tabs **310** may extend upward from the first floor **312**. In some embodiments, the outer surface **324** of each of the plurality of tabs **310** extends at an angle **564** with respect to an axis (e.g., axis **502**) orthogonal to the upper surface **208** of the first plate **204**. In some embodiments, the angle **564** is about 5 degrees to about 20 degrees. In such embodiments, a thickness of the plurality of tabs **310** decreases from a lower portion **504** of the plurality of tabs **310** to an upper portion **506** of the plurality of tabs **310** so that less force or pressure is required to swage the plurality of tabs **310**.

(27) FIG. **5**B depicts a cross-sectional side view of a tab of the plurality of tabs **310** in accordance with at least some embodiments of the present disclosure. The inner surface **322** of each of the plurality of tabs **310** is a support element facing surface. In some embodiments, a base **518** of the inner surface **322** is spaced from the support element **218** by a distance **520**. In some embodiments, the distance **520** is about 0.1 to about 0.8 mm. The spacing between the tab and the support element prevents the base **518** from contacting the support element when the tab is swaged, removing any force or pressure on portions of the support element opposite the base **518**. In such embodiments, when swaged, the plurality of tabs **310** may minimally contact the plurality of support elements **218** at locations corresponding with the upper portion **506** of the plurality of tabs **310**. In some

embodiments, when swaged, the plurality tabs **310** may be bent over but not in direct contact with the plurality of support elements **218**, exerting no force or pressure on the plurality of support elements **218**.
(28) FIG. 5C depicts a cross-sectional side view of a tab of the plurality of tabs **310** in accordance with at least some embodiments of the present disclosure. In some embodiments, a plane of bending **550** of the plurality of tabs **310** is disposed above a horizontal center plane **560** of the plurality of support elements **218**. The plane of bending **550** may corresponding with an interface between the first floor **312** and the lower portion **504** of the plurality of tabs **310**. The plurality of support elements **218** may be more prone to cracking, chipping or breaking when stress or force is exerted along the horizontal center plane **560** of the plurality of support elements **218**. In some embodiments, the plane of bending **550** is disposed vertically above the horizontal center plane **560** to reduce stress or force proximate the horizontal center plane **560** of the plurality of support elements **218**. In some embodiments, the first floor **312** may include an annular recess **562** at an interface between the first floor **312** and the second floor **510**. The annular recess **562** advantageously forms a space between the plurality of support elements **218** along the horizontal center plane **560** and the first plate **204** so that no stress or pressure is exerted on the plurality of support elements **218** proximate the horizontal center plane **560**. In some embodiments, the annular recess **562** has a height **566** of about to about 0.8 mm from the first floor **312** to a lower surface **568** of the annular recess **562**.
(29) FIG. **6** depicts a schematic cross-sectional view of a portion of a first plate **204** in accordance with at least some embodiments of the present disclosure. In some embodiments, the inner surface **322** of the plurality of tabs **310** has a width **606** less than a width **608** of the outer surface **324**. In some embodiments, the second floor **510** is substantially flat.
(30) FIG. **7** depicts a schematic cross-sectional view of a portion of a first plate **204** with a plurality of tabs **310** in a bent position in accordance with at least some embodiments of the present disclosure. As shown in FIG. **7**, in some embodiments, the plurality of tabs **310**, when bent, may minimally contact the plurality of support elements **218** at an upper portion of the plurality of support elements **218**. In some embodiments, the plurality of tabs **310** may be bent over the plurality of support elements **218** without contacting the plurality of support elements **218**.
(31) While the foregoing is directed to embodiments of the present disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof.

Claims

1. A substrate support, comprising: a first plate having a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings; a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate; and a plurality of support elements disposed in corresponding ones of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings.
2. The substrate support of claim 1, wherein the plurality of support elements are ball-shaped.
3. The substrate support of claim 1, wherein the plurality of tabs consist of 4 tabs or 6 tabs.
4. The substrate support of claim 1, wherein an inner surface of each of the plurality of tabs faces a respective support element of the plurality of support elements, and wherein a base of the inner surface is spaced from the respective support element.
5. The substrate support of claim 1, wherein a thickness of the plurality of tabs decrease from a lower portion of the plurality of tabs to an upper portion of the plurality of tabs.
6. The substrate support of claim 1, wherein an outer surface of each of the plurality of tabs extends at an angle of about 5 degrees to about 20 degrees with respect to an axis orthogonal to an upper

surface of the first plate.
7. The substrate support of claim 1, wherein a bottom surface of each respective opening includes a first floor disposed below the plurality of tabs and a second floor extending from the first floor towards a lower surface of the first plate to accommodate the plurality of support elements.
8. The substrate support of claim 1, wherein the plurality of openings are disposed at regular intervals along circles that are concentric with a center of the first plate.
9. The substrate support of claim 1, wherein the plurality of tabs include a lower portion and an upper portion, wherein the lower portion is thicker than the upper portion.
10. A substrate support, comprising: a first plate comprising: a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings; a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate; and a plurality of support elements disposed in corresponding ones of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings; a second plate coupled to a lower surface of the first plate, the second plate having one or more heater elements; and a support shaft coupled to the first plate and the second plate.
11. The substrate support of claim 10, wherein the one or more heater elements comprise one or more inner heater elements and one or more outer heater elements.
12. The substrate support of claim 11, wherein the second plate includes a thermal break between the one or more inner heater elements and one or more outer heater elements.
13. The substrate support of claim 10, wherein the plurality of tabs include 4 or more tabs.
14. The substrate support of claim 10, wherein a plane of bending of the plurality of tabs is disposed above a horizontal center plane of the plurality of support elements.
15. The substrate support of claim 10, wherein the plurality of support elements extend above the upper surface of the first plate.
16. The substrate support of claim 10, wherein an inner surface of the plurality of tabs has a width less a width of an outer surface of the plurality of tabs.
17. A degas chamber, comprising: a chamber body defining an interior volume therein; a substrate support for supporting a substrate disposed in the interior volume, the substrate support comprising: a pedestal having a first plate having a plurality of openings extending into the first plate from an upper surface of the first plate to a bottom surface of each respective opening of the plurality of openings, a plurality of tabs comprising three or more tabs disposed in each respective opening and extending towards the upper surface of the first plate, and a plurality of support elements disposed in corresponding ones of the plurality of openings, wherein the plurality of tabs are disposed around each of the plurality of support elements and configured to retain the plurality of support elements in the plurality of openings; and a support shaft coupled to the pedestal; and a heating lamp disposed in the interior volume opposite the substrate support.
18. The degas chamber of claim 17, wherein the pedestal includes a second plate coupled to the first plate, wherein the second plate includes one or more resistive heating elements disposed therein.
19. The degas chamber of claim 17, wherein a bending plane of the plurality of tabs is disposed above a horizontal center plane of the plurality of support elements.
20. The degas chamber of claim 17, further comprising a gas supply coupled to the chamber body to supply one or more process gases to the interior volume.